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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » Data Science for Engineers (course)

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Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

Week 3 ()

Week 4 ()

Week 5 ()

Week 6 ()

Module : Predictive Modelling (unit?

Week 6 : Assignment 6

The due date for submitting this assignment has passed.

Due on 2025-09-03, 23:59 IST.

Assignment submitted on 2025-09-02, 13:33 IST

For the following set of questions 1, 2, 3, 4, 5, use the dataset **bonds.txt.** (https://drive.google.com/file/d/1jNVN6m9LA6D5x0jXbI3WldIk0_uop5-v/view?usp=sharing)This dataset contains 2 variables, Coupon rate and Bid price.

1) What is the relationship between the variables, Coupon rate and Bid price? ***1 point***

- Coupon rate = 99.95 + 0.24 * Bid price
- Bid price = 99.95 + 0.24 * Coupon rate
- Bid price = 74.7865 + 3.066 * Coupon rate
- Coupon rate = 74.7865 + 3.066 * Bid price

Yes, the answer is correct.
Score: 1

Accepted Answers:
*Bid price = 74.7865 + 3.066 * Coupon rate*

2) Choose the correct option that best describes the relation between the variables Coupon rate and Bid price in the given data. ***1 point***

- Strong positive correlation
- Weak positive correlation
- Strong negative correlation
- Weak negative correlation

Yes, the answer is correct.
Score: 1

Accepted Answers:
Strong positive correlation

unit=72&lesson=73)

- Linear Regression (unit?unit=72&lesson=74)
- Model Assessment (unit?unit=72&lesson=75)
- Diagnostics to Improve Linear Model Fit (unit?unit=72&lesson=76)
- Simple Linear Regression Model Building (unit?unit=72&lesson=77)
- Simple Linear Regression Model Assessment (unit?unit=72&lesson=78)
- Simple Linear Regression Model Assessment (Continued) (unit?unit=72&lesson=79)
- Muliple Linear Regression (unit?unit=72&lesson=80)
- Dataset (unit?unit=72&lesson=81)
- FAQ (unit?unit=72&lesson=82)

3) What is the R-Squared value of the model obtained in Q1? **1 point**

- 0.2413
- 0.12
- 0.7516
- 0.5

Yes, the answer is correct.
Score: 1

Accepted Answers:
0.7516

4) What is the adjusted R-Squared value of the model obtained in Q1? **1 point**

- 0.22
- 0.7441
- 0.088
- 0.5

Yes, the answer is correct.
Score: 1

Accepted Answers:
0.7441

5) Based on the model relationship obtained from Q1, what is the residual error obtained while calculating the bid price of a bond with coupon rate of 3? **1 point**

- 10.5155
- -10.5155
- 6.17
- 0

Yes, the answer is correct.
Score: 1

Accepted Answers:
10.5155

6) State whether the following statement is True or False. **1 point**
Covariance is a better metric to analyze the association between two numerical variables than correlation.

- True
- False

Yes, the answer is correct.
Score: 1

Accepted Answers:
False

7) If R^2 is 0.4, SSR=200 and SST=500, then SSE is **1 point**

- 500
- 200

Week 6 Feedback Form : Data Science for Engineers!! (unit?unit=72&lesson=158)

Quiz: Week 6 : Assignment 6 (assessment?name=241)

Week 7 ()

Text Transcripts ()

Download Videos ()

Books ()

Problem Solving Session - July 2025 ()

300

None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
300

8) Linear Regression is an optimization problem where we attempt to minimize ***1 point***

SSR (residual sum-of-squares)

SST (total sum-of-squares)

SSE (sum-squared error)

Slope

Yes, the answer is correct.
Score: 1

Accepted Answers:
SSE (sum-squared error)

9) The model built from the data given below is $Y = 0.2x + 60$. Find the values for R^2 and Adjusted R^2 . ***1 point***

X	80	75	85	70	65
Y	85	70	80	95	70

Table 1.2.Q

R^2 is 0.22 and Adjusted R^2 is −0.303

R^2 is 0.022 and Adjusted R^2 is −0.303

R^2 is 0.022 and Adjusted R^2 is 0.303

None of the above

No, the answer is incorrect.
Score: 0

Accepted Answers:
R^2 is 0.22 and Adjusted R^2 is −0.303

10) Identify the parameters β_0 and β_1 that fits the linear model $\beta_0 + \beta_1 x$ using the following information: total sum of squares of $X, SS_{XX} = 52.53, SS_{XY} = 52.01,$ mean of $X, \overline{X} = 4.46,$ and mean of $Y, \hat{Y} = 6.32.$ ***1 point***

1.9 and 0.99

10.74 and 1.01

4.42 and 1.01

None of the above

Yes, the answer is correct.
Score: 1

Accepted Answers:
1.9 and 0.99